

A19 Maxilerator

Parameterizable Output-Stationary INT8 Systolic Array AI Accelerator

Discord Name	Affiliation	Role	Experience
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muhammedrabin	Technical University of Munich	Team Member (Output Processor)	Master's Student
myx_on_57134	National Taiwan University of Science and Technology	Team Member (Memory)	Master's Student
maroonchocopie	Indian Institute of Technology, Delhi	Team Member (MAC, Systolic Array)	Master's Student

Video Link: <https://drive.google.com/file/d/1-m7rP2tFc2djv29NrM6ZzOCDwGAntJ3e/view?usp=sharing>

Project Information, Design Objectives & Constraints

Project Context

- SSCS Chipathon 2026 — **Track A: Foundational Building Blocks**
- Target PDK: **GF180MCU**
- Target slot: **Slot A**, approx. 1.25 mm²

What We Are Building

This project implements a compact **8×8 output-stationary systolic array accelerator** for tiled INT8 matrix multiplication, targeting TinyML / CNN-style edge inference workloads. The design is structured as a reusable digital IP block rather than only a one-time chip demo.

“The goal is not only to build an accelerator, but to package the systolic-array compute engine as a reusable, parameterizable, and verification-ready digital IP block.”

What Makes This Design Stand Out

- **Three-level reusability:** reusable arithmetic primitives, reusable `accelerator_core`, and wrapper-level integration paths
- **Throughput-oriented ping-pong buffering:** overlaps tile loading with computation to improve effective throughput
- **Parameterized architecture:** array size, accumulator width, and interface width can be adapted for different workloads, precision needs, area budgets, and integration targets
- **Output-stationary dataflow:** keeps partial sums local inside MAC units, reducing intermediate data movement
- **Verification-friendly RTL:** clean datapath/control separation and module-level testability

Main Design Objective

Build a compact, reusable, and verifiable **INT8 matrix-multiplication accelerator** suitable for open-source silicon and TinyML-style edge inference workloads.

Functional Objectives

- Implement an **8×8 output-stationary systolic array**
- Support signed **INT8 × INT8 multiplication**
- Use **21-bit accumulation** for safe partial-sum storage
- Support tiled matrix multiplication for matrices larger than 8×8
- Use two **128×8 SRAM macros** in ping-pong mode
- Target **25 MHz** operation on GF180MCU
- Fit within **Slot A** area constraints

RTL / Reusability Objectives

- Keep `mac_unit` and `systolic_array` reusable as arithmetic building blocks
- Provide `accelerator_core` as the main reusable IP boundary
- Separate the Chipathon physical interface from the reusable accelerator logic
- Support wrapper-level integration, including an AXI4-Stream reference path
- Keep datapath, memory, controller, and output logic clearly separated

Design Constraints

- Slot A area budget: approx. 1.25 mm²
- 22 physical-facing top-level pins
- Target clock: 25 MHz
- SRAM macros are single-port and 8-bit wide
- Must support tiled computation beyond native 8×8 array size

System Overview

RTL System Overview

- **accelerator_top**: chip-facing wrapper with 22 pins
- **host_interface**: translates pad protocol to core signals
- **accelerator_core**: reusable compute/control IP boundary
- **memory**: two 128x8 SRAMs for ping-pong buffering
- **feeder**: generates skewed A/B streams
- **systolic_array**: 8x8 grid of 64 reusable mac_units
- **output_processor**: serializes saturated INT8 results

Tile-Level Processing Flow

1. Host sends 128-byte tile
2. Feeder creates skewed A/B streams from SRAM
3. Systolic array performs local MAC accumulation
4. Next tile loads into inactive SRAM in parallel
5. Controller swaps SRAM roles after tile_done
6. On last_pass, feeder drains remaining wavefront
7. Output processor serializes 64 INT8 results

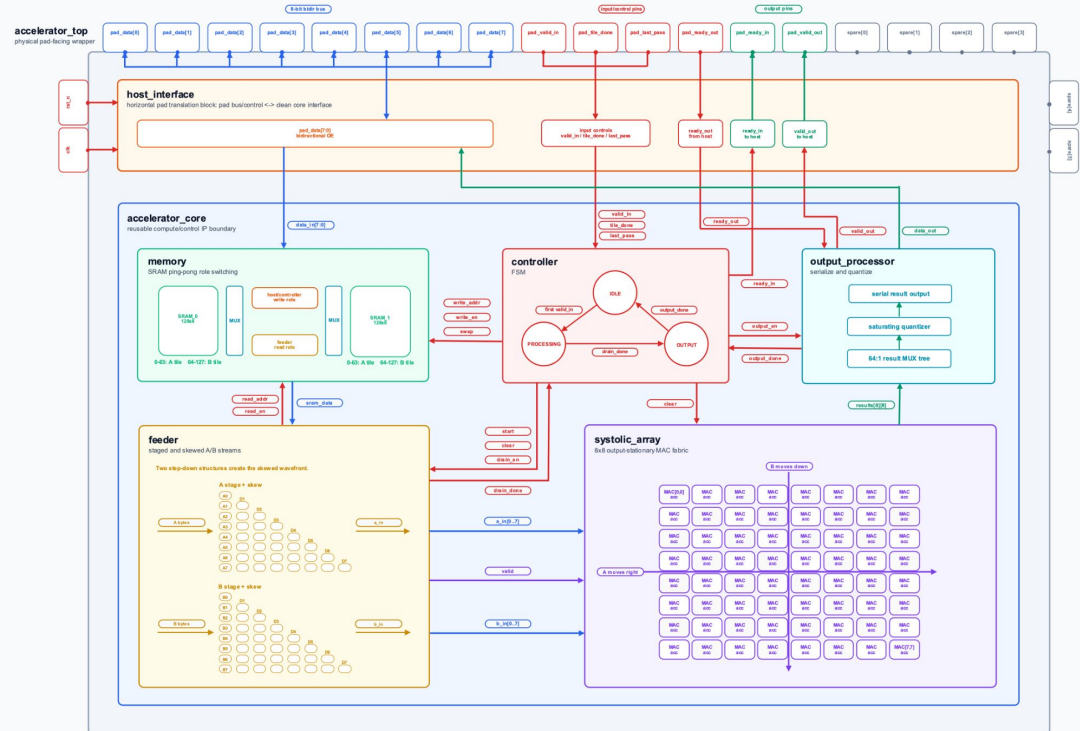
Key RTL Idea

- Data path: memory → feeder → systolic_array → output_processor
- Control path: controller FSM → write_en / swap / clear / output_en
- Ping-pong buffering overlaps tile load and compute
- Output-stationary flow keeps partial sums inside MACs

Team Maxilerator - Parameterizable Output-Stationary INT8 Systolic Array AI Accelerator - Schematic

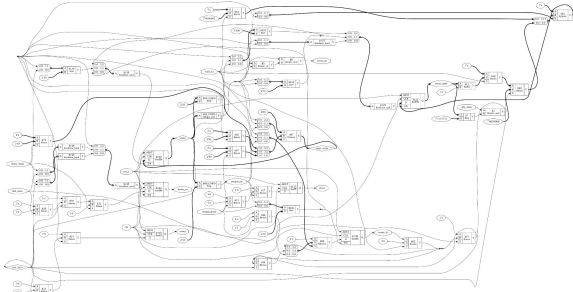
22 top-level pins -> host_interface -> accelerator_core floorplan

data control result compute wavefront

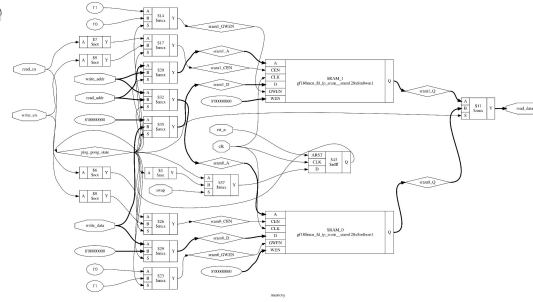


RTL Schematics

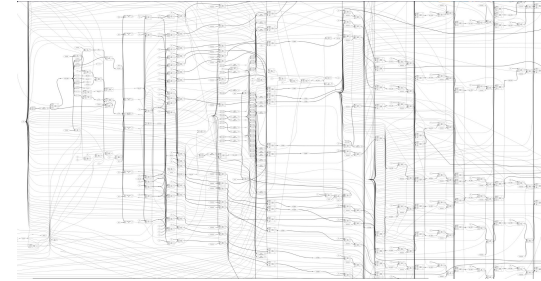
controller.sv



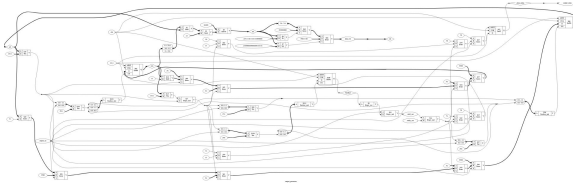
memory.sv



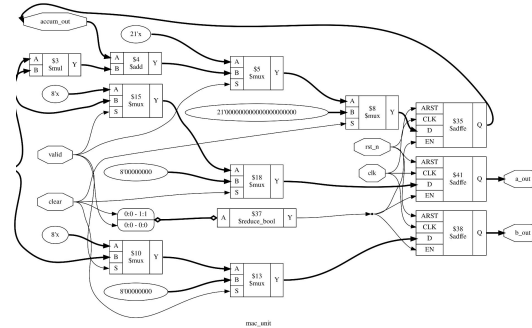
feeder.sv



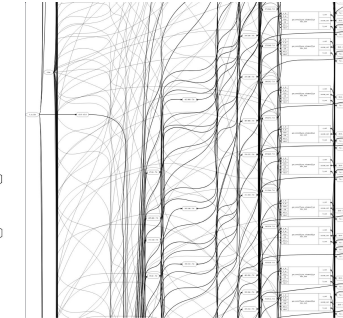
output_processor.sv



mac_unit.sv



systolic_array.sv



Simulation Results

controller.sv

```
73773.01ns INFO cocotb.controller PASS: test_single_cycle_pulses
73773.01ns INFO cocotb.regression tb_controller.test_single_cycle_pulses passed
73773.01ns INFO cocotb.regression

** TEST STATUS SIM TIME (ns) REAL TIME (s) RATIO (ns/s) **
*****
** tb_controller.test_reset PASS 80.00 0.02 5135.75 **
** tb_controller.test_idle_to_processing PASS 161.00 0.00 125121.91 **
** tb_controller.test_write_addr_increment PASS 601.00 0.00 285188.00 **
** tb_controller.test_surplus_tile_done PASS 561.00 0.00 365316.65 **
** tb_controller.test_non_last_tile_swap PASS 5281.00 0.01 385608.43 **
** tb_controller.test_last_tile PASS 5281.00 0.01 446026.45 **
** tb_controller.test_drain_done_to_output PASS 5841.00 0.01 552698.86 **
** tb_controller.test_output_phase PASS 5361.00 0.01 476805.36 **
** tb_controller.test_single_pass PASS 8441.00 0.01 527366.70 **
** tb_controller.test_multi_pass PASS 23801.00 0.05 521788.78 **
** tb_controller.test_host_stall PASS 2561.00 0.01 452507.06 **
** tb_controller.test_write_en_gating PASS 5282.00 0.01 551445.27 **
** tb_controller.test_single_cycle_pulses PASS 10521.00 0.03 349472.74 **
*****
** TESTS=13 PASS=13 FAIL=0 SKIP=0 73773.01 0.19 378956.65 **
*****

make[1]: Leaving directory '/foss/designs/test_repo/Systolic-Tensor-Accelerator/cocotb'
```

feeder.sv

```
(PASS) test_back_to_back_tiles
93815.01ns INFO cocotb.regression tb_feeder.test_back_to_back_tiles passed
93815.01ns INFO cocotb.regression

** TEST STATUS SIM TIME (ns) REAL TIME (s) RATIO (ns/s) **
*****
** tb_feeder.test_reset_state PASS 121.00 0.01 9601.04 **
** tb_feeder.test_no_autorstart PASS 9631.00 0.01 159645.36 **
** tb_feeder.test_address_sequence PASS 5481.00 0.04 154450.16 **
** tb_feeder.test_read_en_self_stop PASS 5681.00 0.03 203583.68 **
** tb_feeder.test_valid_pulse_timing PASS 6481.00 0.03 222890.54 **
** tb_feeder.test_sawtooth_normal PASS 6481.00 0.03 212777.06 **
** tb_feeder.test_drain_phase PASS 6441.00 0.03 212377.75 **
** tb_feeder.test_drain_done_timing PASS 6441.00 0.03 198015.94 **
** tb_feeder.test_start_restart PASS 11841.00 0.06 213548.47 **
** tb_feeder.test_clear_full_reset PASS 3881.00 0.01 216889.74 **
** tb_feeder.test_two_tile_continuous PASS 11481.00 0.05 221126.90 **
** tb_feeder.test_valid_no_glitch PASS 6881.00 0.03 185298.25 **
** tb_feeder.test_staging_capture_timing PASS 5881.00 0.04 167977.81 **
** tb_feeder.test_staging_phase_separation PASS 5681.00 0.02 262539.70 **
** tb_feeder.test_back_to_back_tiles PASS 11761.00 0.06 213122.88 **
*****
** TESTS=15 PASS=15 FAIL=0 SKIP=0 93815.01 0.50 189421.68 **
*****

make[1]: Leaving directory '/foss/designs/test_repo/Systolic-Tensor-Accelerator/cocotb'
```

memory.sv

```
32770.01ns INFO cocotb.memory PASS: back_to_back_tiles
32770.01ns INFO cocotb.regression tb_memory.test_back_to_back_tiles passed
32770.01ns INFO cocotb.regression

** TEST STATUS SIM TIME (ns) REAL TIME (s) RATIO (ns/s) **
*****
** tb_memory.test_reset PASS 280.00 0.02 17945.47 **
** tb_memory.test_write_then_read PASS 1481.00 0.00 325853.07 **
** tb_memory.test_read_latency PASS 600.00 0.00 260461.85 **
** tb_memory.test_write_en_gating PASS 641.00 0.00 367237.93 **
** tb_memory.test_ping_pong_swap PASS 4561.00 0.01 328847.65 **
** tb_memory.test_simultaneous_read_write PASS 801.00 0.00 472802.00 **
** tb_memory.test_full_tile_128_bytes PASS 15841.00 0.06 281148.90 **
** tb_memory.test_swap_preserves_data PASS 1441.00 0.00 322837.09 **
** tb_memory.test_read_en_gating PASS 801.00 0.00 428112.23 **
** tb_memory.test_multiple_swaps PASS 1041.00 0.00 408865.11 **
** tb_memory.test_address_boundaries PASS 721.00 0.00 483777.51 **
** tb_memory.test_back_to_back_tiles PASS 4481.00 0.01 387040.28 **
*****
** TESTS=12 PASS=12 FAIL=0 SKIP=0 32770.01 0.14 236214.22 **
*****

make[1]: Leaving directory '/foss/designs/test_repo/Systolic-Tensor-Accelerator/cocotb'
```

output_processor.sv

```
42880.01ns INFO cocotb.output_processor PASS: random_with_saturation (5 trials)
42880.01ns INFO cocotb.regression tb_output_processor.test_random_with_saturation passed
42880.01ns INFO cocotb.regression

** TEST STATUS SIM TIME (ns) REAL TIME (s) RATIO (ns/s) **
*****
** tb_output_processor.test_reset PASS 120.00 0.02 5496.76 **
** tb_output_processor.test_valid_out_gated_by_output_en PASS 240.00 0.00 139133.79 **
** tb_output_processor.test_serialization_order PASS 2800.00 0.01 225513.21 **
** tb_output_processor.test_saturation_positive PASS 2800.00 0.02 140465.34 **
** tb_output_processor.test_saturation_negative PASS 2800.00 0.01 337977.76 **
** tb_output_processor.test_saturation_boundary PASS 2800.00 0.01 382979.81 **
** tb_output_processor.test_output_done_single_pulse PASS 3120.00 0.01 448373.48 **
** tb_output_processor.test_backpressure PASS 3000.00 0.01 408877.58 **
** tb_output_processor.test_backpressure_intermittent PASS 5360.00 0.01 427839.91 **
** tb_output_processor.test_output_en_reset PASS 3000.00 0.01 309104.69 **
** tb_output_processor.test_random_with_saturation PASS 14000.00 0.05 298833.83 **
*****
** TESTS=12 PASS=12 FAIL=0 SKIP=0 42880.01 0.22 197835.00 **
*****

make[1]: Leaving directory '/foss/designs/test_repo/Systolic-Tensor-Accelerator/cocotb'
```

Simulation Results (contd.)

mac_unit.sv

```

25281.01ns INFO cocotb.mac_unit PASS: clear then reaccumulate
25281.01ns INFO cocotb.regression tb mac_unit.test_clear then reaccumulate passed
25281.01ns INFO cocotb.regression
*****
** TEST STATUS SIM TIME (ns) REAL TIME (s) RATIO (ns/s) **
*****
** tb_mac_unit.test_reset PASS 80.00 0.01 6625.68 **
** tb_mac_unit.test_single_accumulate PASS 1320.00 0.00 277963.72 **
** tb_mac_unit.test_accumulate_multiple PASS 360.00 0.00 88110.49 **
** tb_mac_unit.test_clear_priority PASS 400.00 0.00 181945.73 **
** tb_mac_unit.test_clear_resets PASS 280.00 0.00 280555.45 **
** tb_mac_unit.test_ab_propagate_always PASS 280.00 0.00 105556.91 **
** tb_mac_unit.test_time_cycle_latency PASS 161.00 0.00 163546.37 **
** tb_mac_unit.test_worst_case_signed PASS 200.00 0.00 272880.00 **
** tb_mac_unit.test_max_accumulation PASS 1440.00 0.00 459485.02 **
** tb_mac_unit.test_no_accum_without_valid PASS 360.00 0.00 359768.75 **
** tb_mac_unit.test_random_stress PASS 20000.00 0.04 466557.36 **
** tb_mac_unit.test_clear_then_reaccumulate PASS 480.00 0.00 366314.76 **
*****
** TESTS=12 PASS=12 FAIL=0 SKIP=0 25281.01 0.10 261183.31 **
*****
make[1]: Leaving directory '/foss/designs/test_repo/Systolic-Tensor-Accelerator/cocotb'

```

systolic_array.sv

```

8360.01ns INFO cocotb.systolic_array PASS: pipeline_flow
8360.01ns INFO cocotb.regression tb systolic_array.test_pipeline_flow passed
8360.01ns INFO cocotb.regression
*****
** TEST STATUS SIM TIME (ns) REAL TIME (s) RATIO (ns/s) **
*****
** tb_systolic_array.test_reset PASS 120.00 0.04 3378.59 **
** tb_systolic_array.test_identity PASS 1080.00 0.01 127325.19 **
** tb_systolic_array.test_all_ones PASS 1080.00 0.01 125889.01 **
** tb_systolic_array.test_single_element PASS 1080.00 0.01 163573.77 **
** tb_systolic_array.test_clear PASS 1160.00 0.01 95766.02 **
** tb_systolic_array.test_signed_negative PASS 1080.00 0.01 99697.34 **
** tb_systolic_array.test_worst_case PASS 1080.00 0.01 94197.18 **
** tb_systolic_array.test_no_accum_without_valid PASS 600.00 0.00 292116.36 **
** tb_systolic_array.test_pipeline_flow PASS 1080.00 0.01 147598.52 **
*****
** TESTS=9 PASS=9 FAIL=0 SKIP=0 8360.01 0.12 78812.17 **
*****
make[1]: Leaving directory '/foss/designs/test_repo/Systolic-Tensor-Accelerator/cocotb'

```

RTL Design Decisions & Feasibility Analysis

Key Design Decisions

Design Choice	Reason
Output-stationary	Keeps partial sums local inside MAC units
INT8 precision	Suitable for compact TinyML-style inference
8x8 array size	Balances compute parallelism and Slot A area
21-bit accumulation	Provides safe range for INT8 partial sums
Ping-pong SRAM	Overlaps tile loading with computation
128-byte tile format	Matches 64 A bytes + 64 B bytes per tile
Serialized output	Reduces physical pin count
Separate host interface	Keeps accelerator_core reusable

Handshake Protocol Summary

Signal	Role
valid_in	Host provides a valid input byte
ready_in	Tile-level permission to start a new tile transfer
tile_done	Completion of 128-byte tile transfer
last_pass	Marks the final computation pass
swap	Switches active/inactive SRAM roles
clear	Clears feeder state and MAC accumulators
output_en	Enables output serialization
valid_out	Output byte is valid
ready_out	Host is ready to accept output byte
output_done	All 64 output bytes transferred

Area Feasibility

Component	Estimated Area / Slot Impact
64 MAC units	~808,636 μm^2 / 64.8%
SRAM_0	~116,119 μm^2 / 9.3%
SRAM_1	~116,119 μm^2 / 9.3%
Feeder	~54,062 μm^2 / 4.3%
Output processor	~46,071 μm^2 / 3.7%
Controller + host interface	<1% combined
Total at 70% utilization	~1,143,718 μm^2 / 91.6% Slot A

Accumulator Width Trade-Off

Width	Max K	Coverage	Extra Area	Slot A	Time @25MHz
21	32	ECG / biosignals	0 μm^2	91.6%	0.464 ms
22	64	Gesture / keyword	15.6k μm^2	92.8%	3.177 ms
23	128	Tiny image cls.	31.1k μm^2	94.1%	23.276 ms
24	256	Small speech	46.7k μm^2	95.3%	177.644 ms
25	512	Medium CNN	62.2k μm^2	96.6%	1.387 s
26	1024	Large CNN / ResNet	77.8k μm^2	97.8%	10.958 s
27	2048	BERT-small	93.3k μm^2	99.1%	1.45 min
28	4096	GPT-2 small	108.9k μm^2	100.3%	11.58 min

Project Status, Verification Strategy & Open Questions

Current Project Status

Area	Status
Architecture	Defined and documented
System block diagram	Completed
Module hierarchy	Defined
RTL interfaces	Defined at signal level
Core RTL modules	Implemented and under refinement
Basic verification	Completed for main functional paths
Corner-case testing	Ongoing
Module synthesis	Initial synthesis checks completed
Area estimation	Preliminary estimate prepared
Physical design	Next phase after RTL freeze

Verification Status

- Basic module-level and functional-path tests completed
- cocotb-based verification flow is being expanded
- Python/NumPy reference model used for expected matrix results
- Current focus is **rigorous corner-case testing**
- Verification is being extended to cover fault conditions and protocol edge cases

Questions for Reviewers

- Is **70% utilization** a realistic target for this design in GF180MCU, considering routing congestion, SRAM placement, and the 8x8 MAC array?
- Is the area estimate reasonable for proceeding toward RTL freeze?
- Is the RTL module partitioning suitable for Track A?
- Is the architecture mature enough to move toward synthesis and physical design?

Ongoing / Planned Verification Coverage

Module / Area	Verification Focus
mac_unit	Signed INT8 MAC, accumulator clear/reset
systolic_array	8x8 result correctness and accumulation flow
memory	SRAM read/write and ping-pong role switching
feeder	Skew timing, restart behavior, drain phase
controller	FSM transitions, swap, clear, output control
output_processor	Saturation, serialization, output backpressure
accelerator_core	End-to-end tiled matrix multiplication
protocol_edges	ready_in, tile_done, last_pass, ready_out
fault_cases	reset during operation, invalid/stalled input

End-to-End Test Scenarios

- 8x8 single-tile matrix multiplication
- Multi-tile operation for larger matrices
- Back-to-back tile transfers
- **tile_done** and **last_pass** timing checks
- **ready_in** tile-permission behavior
- Output backpressure using **ready_out**
- Reset/clear during active operation
- Stalled or delayed input transfer